



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Transfer of Thermal Energy

2.3

STUDENT NAME



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PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

2.3 Transfer of thermal energy**2.3.1 Conduction**

- 1 Describe experiments to distinguish between good and bad thermal conductors
- 2 Describe thermal conduction in all solids in terms of atomic or molecular lattice vibrations and also in terms of the movement of free (delocalised) electrons in metallic conductors

2.3.2 Convection

- 1 Explain convection in liquids and gases in terms of density changes and describe experiments to illustrate convection

2.3.3 Radiation

- 1 Describe the process of thermal energy transfer by infrared radiation and know that it does not require a medium
- 2 Describe the effect of surface colour (black or white) and texture (dull or shiny) on the emission, absorption and reflection of infrared radiation
- 3 Describe how the rate of emission of radiation depends on the surface temperature and surface area of an object
- 4 Describe experiments to distinguish between good and bad emitters of infrared radiation
- 5 Describe experiments to distinguish between good and bad absorbers of infrared radiation

2.3.4 Consequences of thermal energy transfer

- 1 Explain everyday applications using ideas about conduction, convection and radiation, including:
 - (a) heating objects such as kitchen pans
 - (b) heating a room by convection
 - (c) measuring temperature using an infrared thermometer
 - (d) using thermal insulation to maintain the temperature of a liquid and to reduce thermal energy transfer in buildings



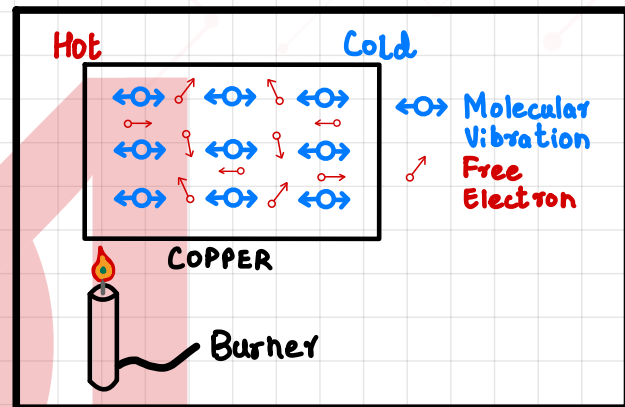
CONDUCTION:

"Mode of transfer of thermal energy done by either molecular vibration or by free electron (efficient) is called conduction."

Conduction is only efficient in solids (metals).

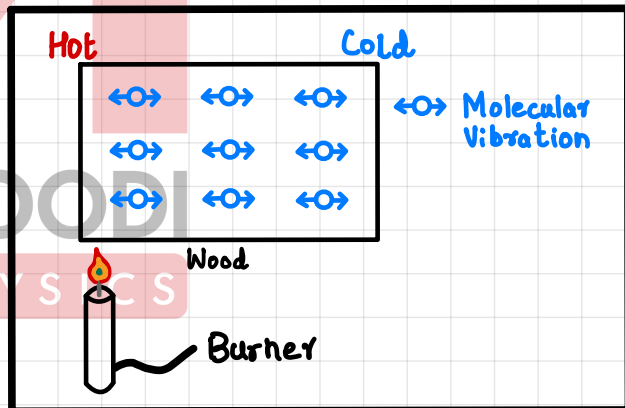
CONDUCTION IN COPPER:

Electrons are free & very small in size as compared to atoms so they can travel hot to cold end efficiently. This is why copper is a good conductor of heat.



CONDUCTION IN WOOD:

Wood stacks energy at hot end, molecules vibrate at hot end vigorously & in turn high amount of heat is generated so wood burns out. Conduction through molecular vibration is less efficient so wood is a bad conductor.



GOOD CONDUCTOR:

"Materials which conducts heat energy efficiently are called good conductors."

They are also known as bad insulators

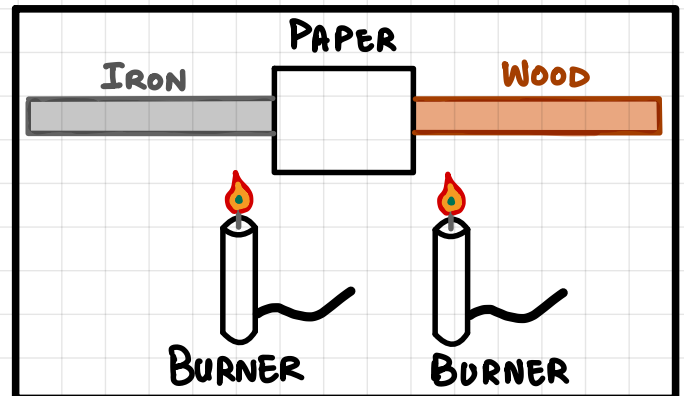
GOOD INSULATOR:

"Materials which conduct heat energy inefficiently are called good insulators."

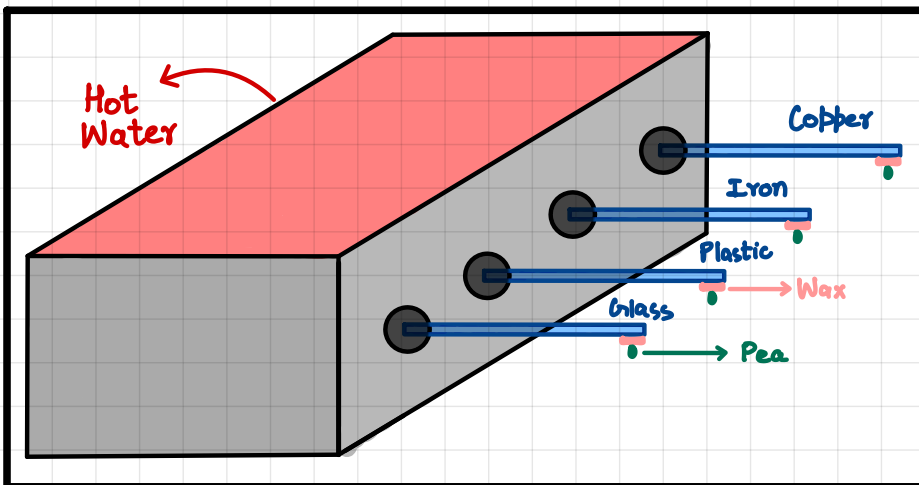
They are also known as bad conductors.

Q Which side of the paper will get burnt?

A Wood side will get burnt as iron circulates heat from hot to cold but wood can't so paper near wood will get burnt.



BEST CONDUCTOR OF HEAT: ***

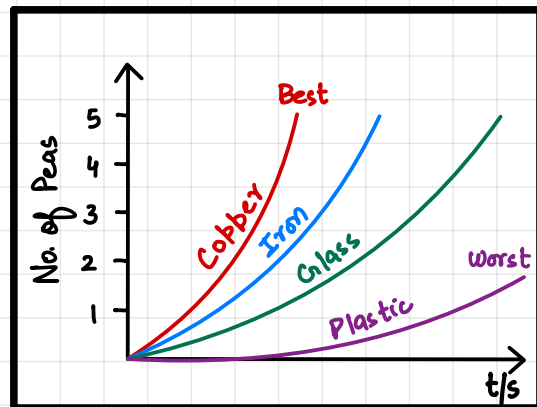
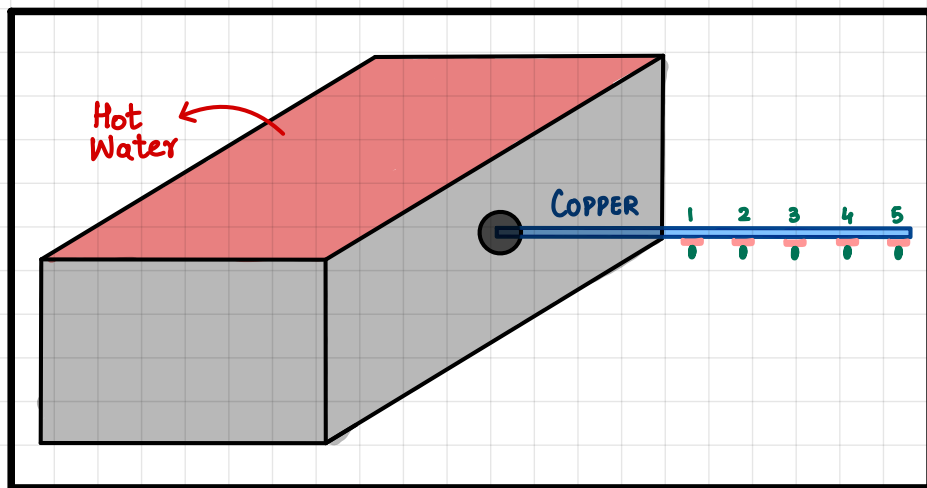


CONSTANT PARAMETERS:

- Length of rods.
- Diameter of rods.
- Size of pea.
- Amount of wax.
- Temperature difference.

The rod whose pea falls first is the best conductor of heat as heat flows fastest in it from hot to cold.

Copper is the best conductor among all.

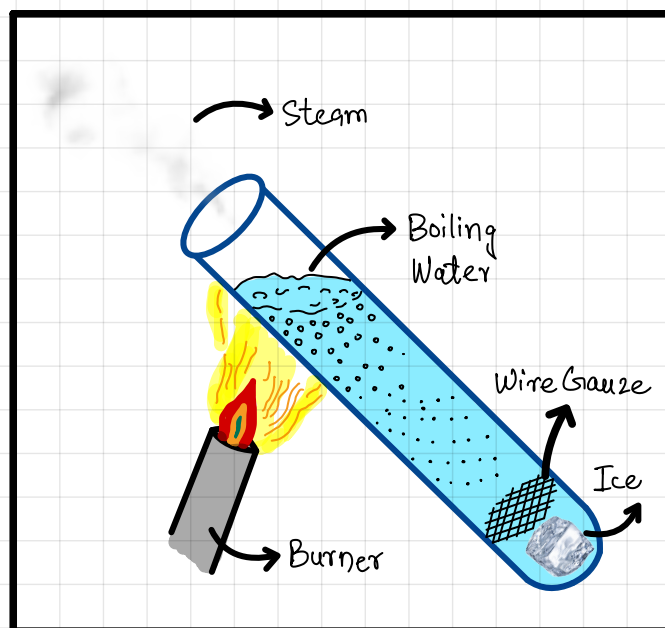


By using multiple peas on one rod, graph of no. of peas fallen vs time can be plotted. High gradient graph represents best conductor.

WATER IS A POOR CONDUCTOR OF HEAT:

If water was a good conductor of heat, it transfers energy from bottom to top as well as from top to bottom. But here if ice is heated from bottom, it melts & if ice is heated from top, it doesn't melt which proves that water is a poor conductor of heat.

Ice is trapped with wire gauze at the bottom to ensure that ice doesn't float on surface.



Ice has less density than water that's why it will rise to the surface of water. It is trapped at the bottom using wire gauze.

Since gas doesn't have free electrons & their molecules are also far apart, they are considered as bad conductor of heat.

CONVECTION:

"Mode of transfer of energy by movement of molecules in a bulk quantity is called convection."

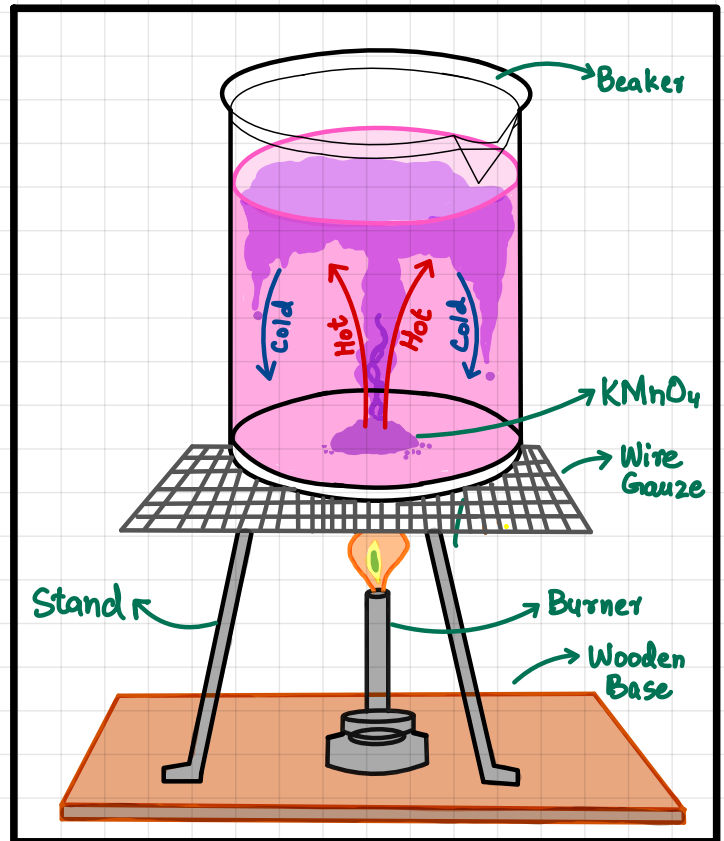
Convection is only efficient in Fluids (Liquids & gases).

CONVECTION IN LIQUIDS

Molecules at the base receive heat, become less dense & rises to the top. Cold water which is denser sinks to the bottom. This process continues until thermal equilibrium is established.

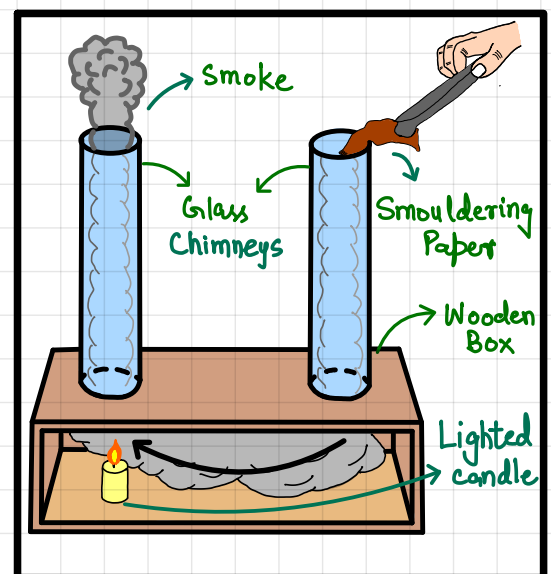
Movement of molecules in this way is called convection current

Potassium permanganate is a purple coloured salt which is used to visualize convection current.



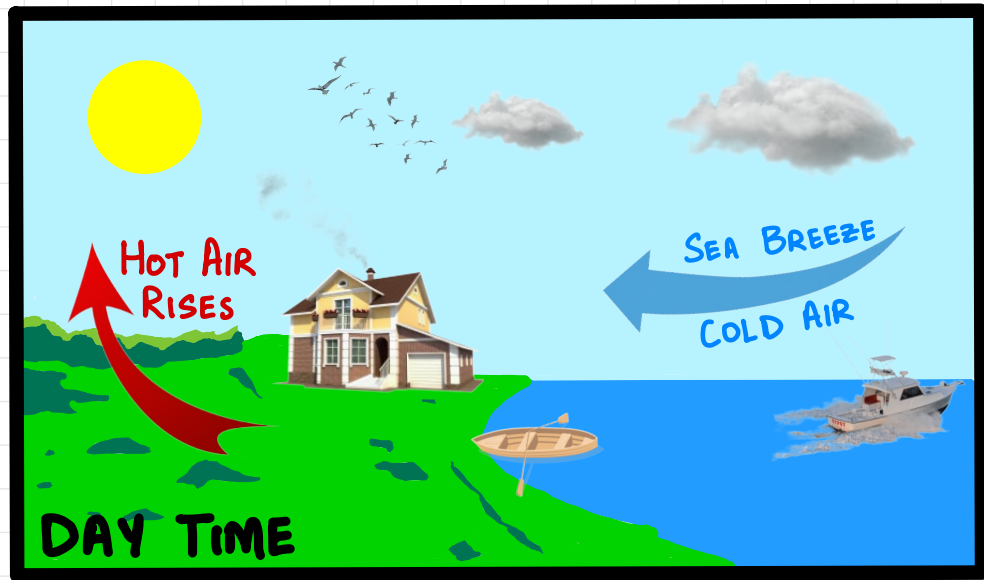
CONVECTION IN GASES:

To observe convection in gases, we place smouldering paper at one end of the chimney & smoke is seen falling down. When smoke reaches near candle, its temperature increases, smoke becomes less dense & rises through other chimney.



SEA BREEZE:

During the day time the land is hotter than sea, so air above the land is hot, becomes less dense & rises while cold air



from the sea replaces it. Since the direction of breeze is from sea to land so it is called sea breeze.

LAND BREEZE:

During the night time the land is cooler than sea, so air above the sea is hot, becomes less dense & rises while cold air



from the land replaces it. Since the direction of breeze is from land to sea so it is called land breeze.

Solid reacts quickly on temperature changes than liquids. So it heats up as well as cool down more quickly.

RADIATION:

"Mode of transfer of energy which doesn't require any medium to travel."

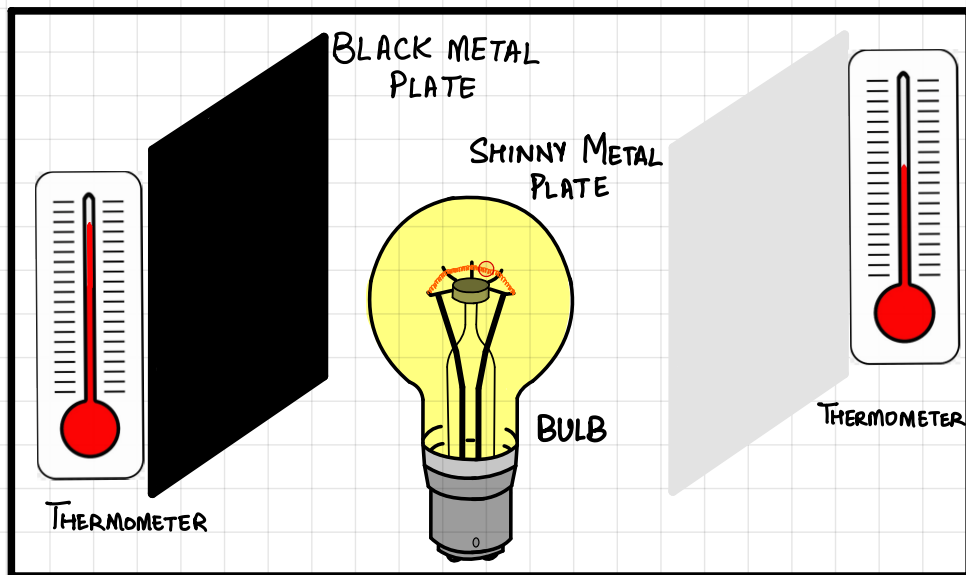
Heat from sun travels to earth as infrared radiations without any aid of the medium.

ABSORPTION, EMISSION & REFLECTION OF RADIATION:

CONSTANT PARAMETERS:

- Size of both plates.
- Distance from bulb.

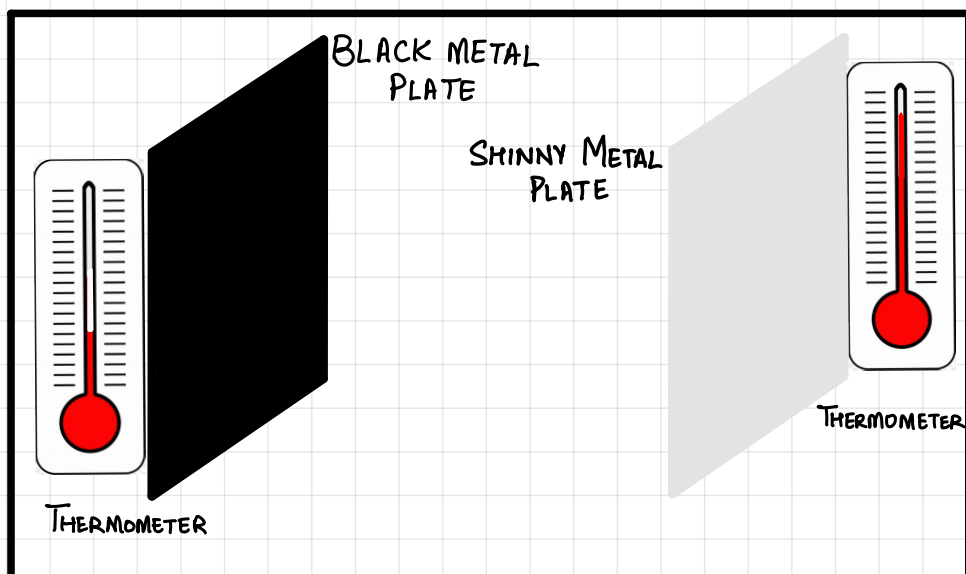
Set the apparatus as shown & start the stop watch. Observe the temperature of thermometer after 30s. It will be observed that black metal plate has



more temperature rise than shiny one which concludes that:

- Black is good absorber of heat.

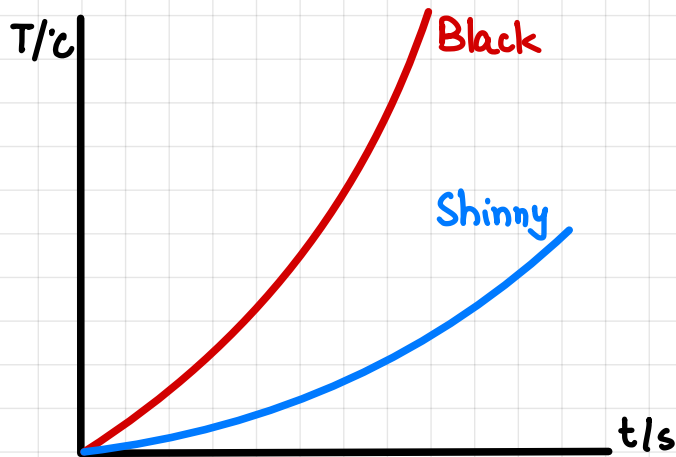
Set the apparatus as shown & start the stop watch. Observe the temperature of thermometer after 30s. It will be observed that black metal plate has less



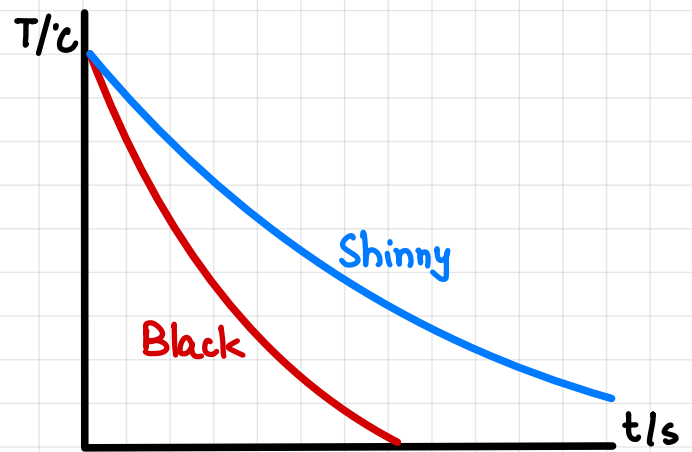
temperature rise then shiny one which concludes that:

- Black is good emitter of heat.

GRAPHS:



HEATING



COOLING

GOOD EMITTER	GOOD ABSORBER	GOOD RELECTOR
BLACK / DULL	BLACK / DULL	SHINY / WHITE

BAD EMITTER	BAD ABSORBER	BAD RELECTOR
SHINY / WHITE	SHINY / WHITE	BLACK / DULL

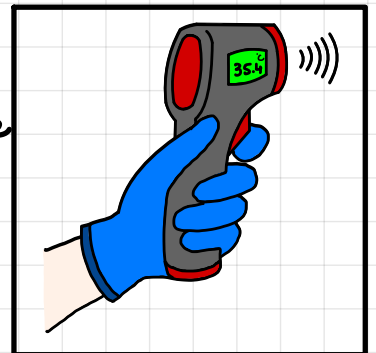
2.3.4

Consequences of Thermal Energy Transfer

EVERYDAY APPLICATIONS:

INFRARED THERMOMETER.

Infrared thermometer uses infrared radiations to measure temperature of body without making physical contact with the body.



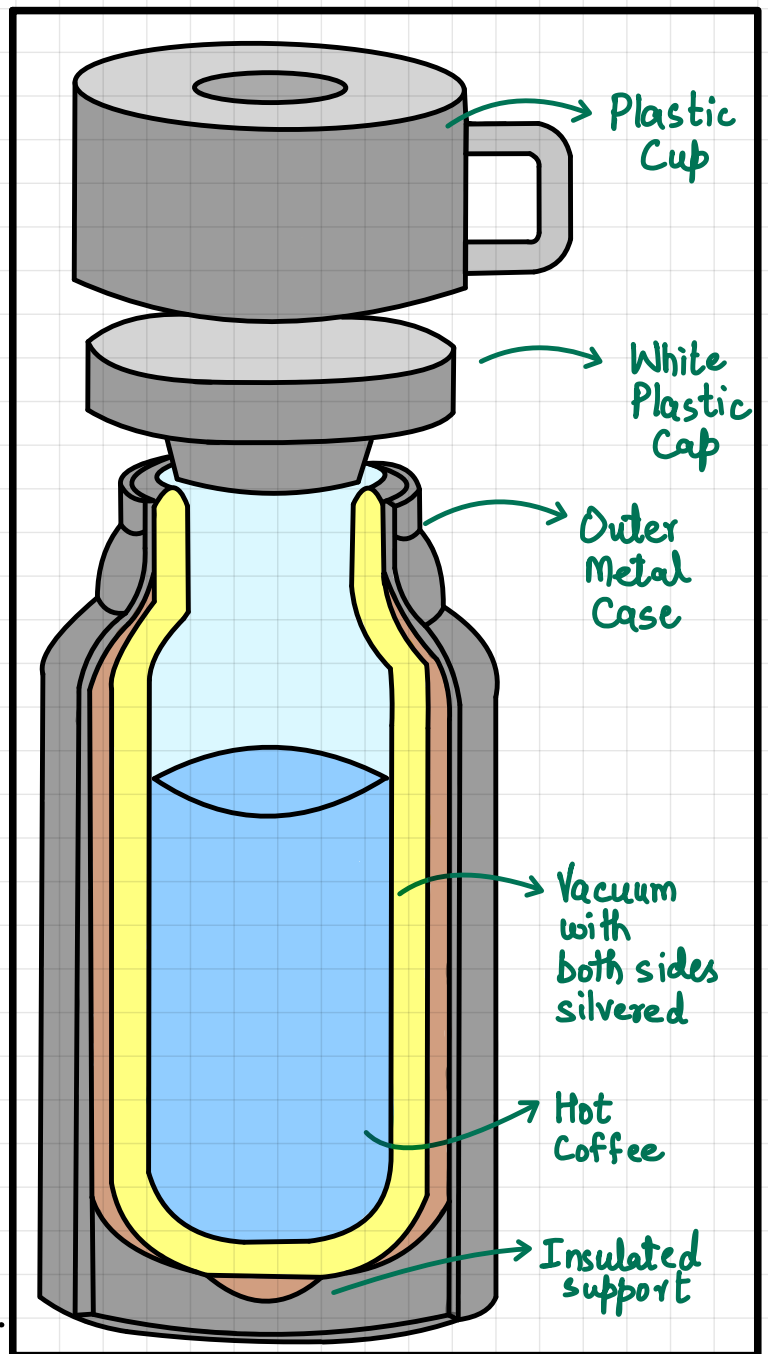
● VACUUM FLASK:

Vacuum Flask reduces heat loss by conduction, convection radiation & evaporation.

Since there is no medium between glass walls so no conduction & convection can't happen as both require medium. Since walls are shiny so heat doesn't flow through as shiny reflects heat.

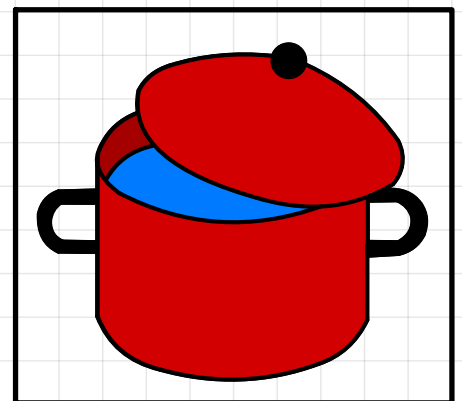
Cap is made from plastic which reduces heat loss by conduction. Since cap is closed so no heat is lost by convection & evaporation.

While white colour reduces heat loss through cap by radiation



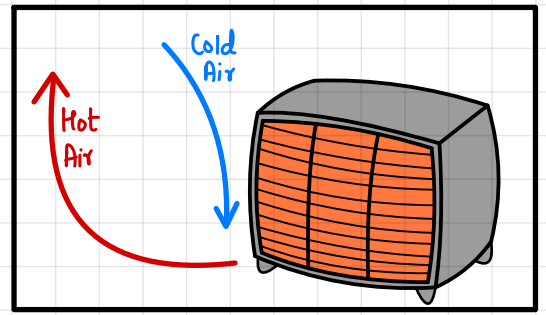
● METAL UTENSILS:

Metal utensils are used in kitchen to cook food as metal are good conductor of heat while the handles are made up of wood or plastic to prevent skin burns.



● ROOM HEATER:

Room Heater makes the room hot by convection of air around it. Air near the heater becomes less dense & rises while cold air sinks to the bottom



INSULATING A HOUSE FROM HEAT:

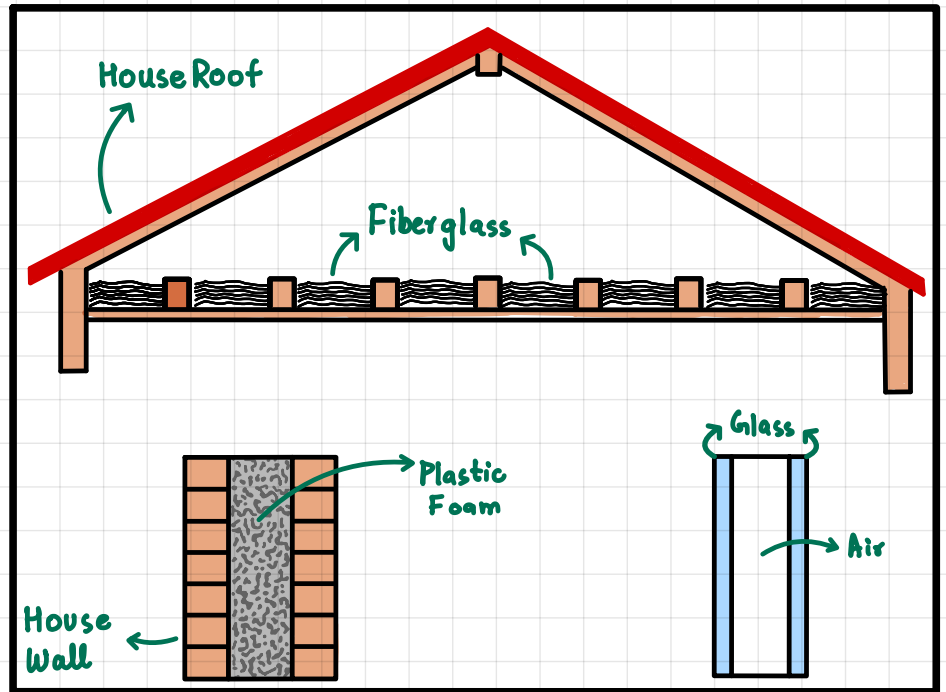
● WALLS:

Heat loss through wall is reduced by putting plastic foam between two walls as foam contains packets of air & air is a bad conductor of heat.

● WINDOWS:

Double glazed windows

reduces heat loss as they contain packets of air between two glass walls & air is a bad conductor of heat.



● ROOF:

Fibre glass on roof reduces heat loss as it contains packets of air & air is a bad conductor of heat.

● FLOOR:

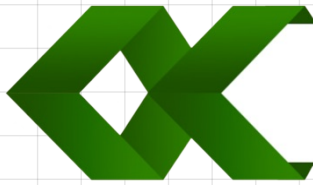
Wooden floors or carpets will reduce heat loss by floor as wood is a bad conductor & carpet contains air, air is a bad conductor of heat.

● CURTAIN:

Using curtains on windows reduces heat loss as it contains air & air is a bad conductor of heat.

● PAINT:

Painting the house white as white is a good reflector of heat



MATCH THE FOLLOWING



Conduction happens due	best conductors of heat.
Metals are the	to molecular vibration & free electrons.
Water is a	to density changes.
Convection is efficient	in liquids & gases.
Convectional happens due	poor conductor of heat
Hot water rises as	require any medium.
Radiation doesn't	it has lower density
Black is the good	reflector of heat.
Shiny is the good	absorber & emitter of heat.
Vacuum Flask reduces	bad conductor of heat as it has packets of air in it.
Fiber glass is a	heat loss by conduction, convection, radiation & evaporation

01 How does heat transfer through a vacuum take place?

- A by conduction, convection and radiation
- B by conduction only
- C by convection only
- D by radiation only

[ON2012/P11/Q15]

02 A rod of metal is heated at one end.

Which statement best describes the conduction of heat through the metal?

- A Atoms move from the hot end and hit electrons at the cold end.
- B Atoms vibrate and hit atoms at the cold end.
- C Free electrons move from the hot end and hit atoms further along the rod.
- D Free electrons vibrate and pass energy to free electrons further along the rod.

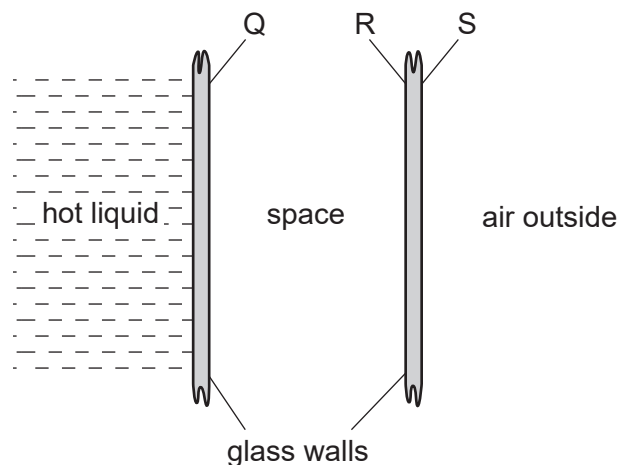
[MJ2012/P11/Q17]

03 How is heat conducted in a metal?

- A by movement of electrons through the metal only
- B by movement of atoms through the metal only
- C by vibration of atoms and movement of electrons through the metal
- D by vibration of atoms only

[MJ2012/P12/Q17]

04 A student uses a double-walled glass vessel to contain a hot liquid.



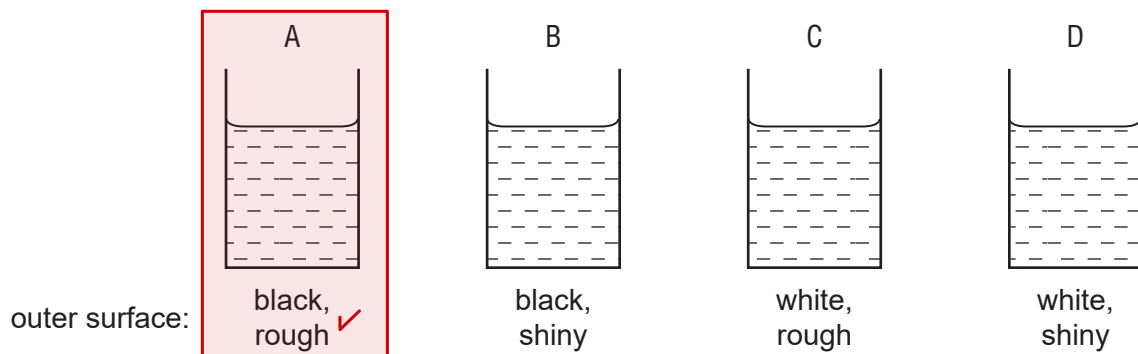
What reduces the heat loss by radiation?

- A a vacuum in the space between the walls
- B painting surface Q black
- C painting surface R black
- D painting surface S silver

[ON2012/P11/Q17]

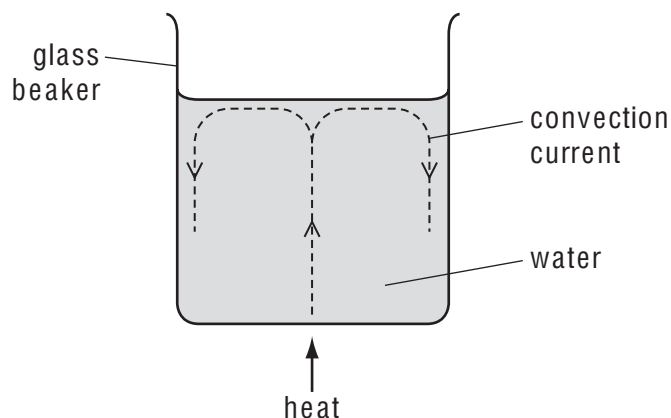
- 05 Four metal cans are identical except for the colour and the texture of their outer surfaces. 100 cm³ of water at 70 °C is poured into each can.

Which cools the most rapidly?



[ON2012/P12/Q17]

- 06 A glass beaker contains water. When the centre of the base of the beaker is heated, a convection current is set up.

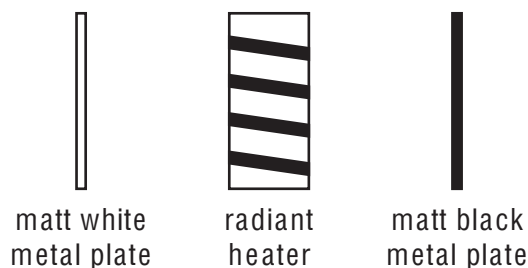


Which statement explains this?

- A The evaporation of water causes water molecules to rise to the surface. ✗
- B The expansion of water molecules causes them to rise to the surface. ✗
- C The water above the heat source rises because it becomes less dense. ✓
- D The water at the sides sinks because it becomes less dense. ✗

[MJ2013/P11/Q13]

- 07 Two identical metal plates are painted, one matt (dull) white and the other matt black. These are placed at equal distances from a radiant heater as shown. The heater is turned on for five minutes.

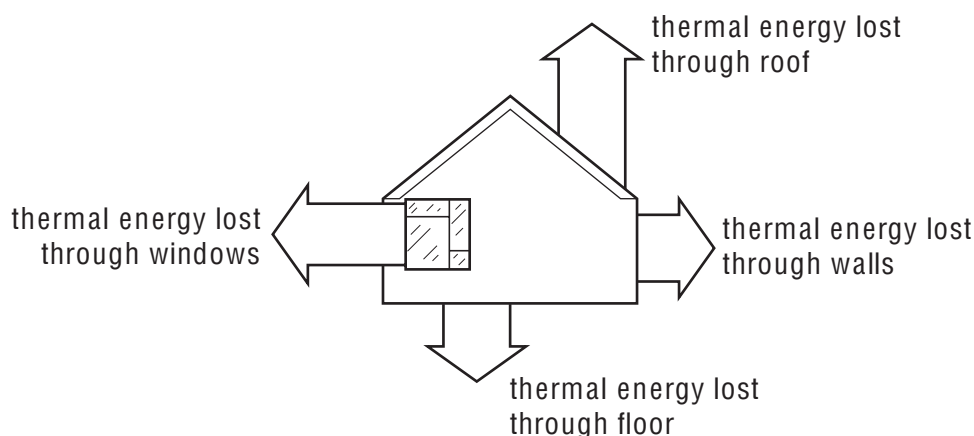


Which metal plate **absorbs more energy** and which plate **emits more energy** in this time?

	absorbs more	emits more
A	black✓	black✓
B	black✓	white
C	white	black✓
D	white	white

[MJ2013/P11/Q14]

- 08 On a cold afternoon, a house loses **54 MJ** of thermal energy (heat) to its surroundings as shown.



The heating system must supply more than **54 MJ** of thermal energy to keep the **temperature** of the **house constant**.

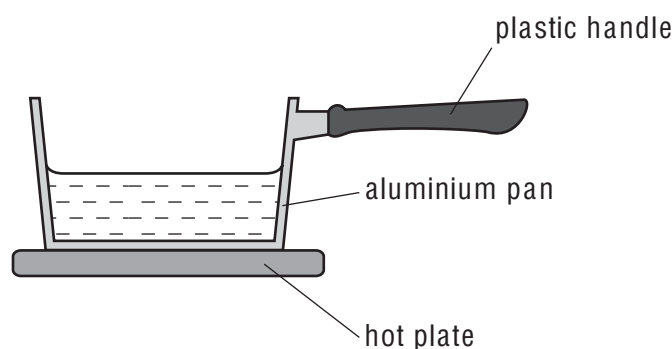
Which statement explains this?

- A** The extra thermal energy is lost from the house to the surroundings by other means.
- B** The extra thermal energy keeps the house warmer than the surroundings.
- C** The temperature of the surroundings decreases continuously during this period.
- D** The thermal insulation of the roof is extremely ineffective.



[ON2013/P11/Q13]

09 A saucepan is used to heat up some water.



How is heat transferred through the aluminium pan and through the plastic handle?

	heat is transferred through the aluminium pan by	heat is transferred through the plastic handle by
A	the movement of free electrons and the vibration of atoms ✓	the movement of free electrons and the vibration of molecules
B	the movement of free electrons and the vibration of atoms ✓	the vibration of molecules only ✓
C	the movement of free electrons only	the movement of free electrons and the vibration of molecules
D	the movement of free electrons only	the vibration of molecules only ✓

[ON2013/P12/Q15]

10 A silver cup is filled with boiling water from a kettle.

A man touches the outside surface of the cup and finds that it is extremely hot.

Why is the surface so hot?

- A Convection takes place in the boiling water. ✗
- B Silver is a good conductor of heat. ✓
- C The boiling water gives out latent heat. ✗
- D The shiny surface is a good emitter of infra-red radiation. ✗

[ON2013/P12/Q16]

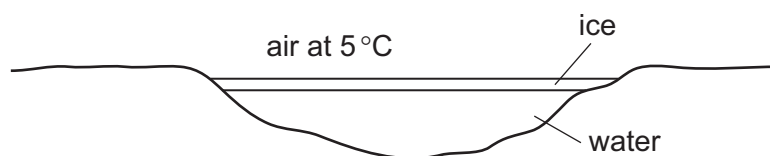
11 A copper rod is heated at one end.

Which statement describes how heat transfer occurs in the copper?

- A Energetic copper molecules ✗ move from the cooler end to the hotter end.
- B Energetic copper molecules ✗ move from the hotter end to the cooler end.
- C Energetic free electrons ✗ move from the cooler end to the hotter end.
- D Energetic free electrons ✓ move from the hotter end to the cooler end. ✓

[ON2014/P11/Q15]

- 12** The diagram shows a frozen pond with the surface of the ice slowly melting as heat is transferred from the warmer air above it.



By which processes is heat transferred from the air to the ice?

- A** conduction, convection and radiation
- B** conduction and convection only
- C** convection and radiation only
- D** radiation and conduction only

[ON2014/P12/Q12]

- 13** A solid bar is heated at one end.

How is thermal energy transferred to the other end of the bar?

- A** Heated molecules move along the bar, carrying energy to the other end.
- B** Heated molecules move along the bar, giving energy to others along the bar.
- C** Heated molecules stay completely still, but give energy to other molecules.
- D** Heated molecules vibrate more rapidly and pass energy to other molecules.

[MJ2015/P11/Q15]

- 14** The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

- A** Black surfaces absorb radiation well.
- B** Black surfaces conduct heat well.
- C** Black surfaces emit radiation well.
- D** Black surfaces reflect radiation well.

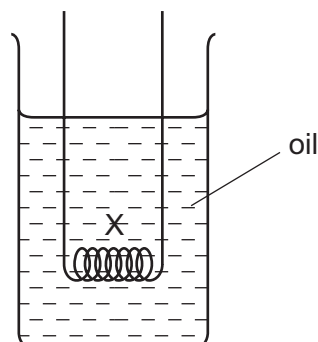
[MJ2015/P11/Q16]

- 15** An electrical heater is placed in a beaker of cold oil, as shown.

The heater is switched on.

What happens to the liquid at X?

- A** It becomes less dense and falls.
- B** It becomes less dense and rises.
- C** It becomes more dense and falls.
- D** It becomes more dense and rises.



[MJ2015/P12/Q15]

16 Which statement is correct?

- A Infra-red radiation cannot travel in a vacuum.
- B Infra-red radiation cannot travel in solids or in gases.
- C Infra-red radiation can only travel in a vacuum.
- D Infra-red radiation can travel in a vacuum and in gases.

[ON2015/P12/Q20]

P1

Mark Scheme

ANSWER KEY			
1	D	11	D
2	C	12	D
3	C	13	D
4	D	14	A
5	A	15	B
6	C	16	D
7	A		
8	A		
9	B		
10	B		

- 1 Fig. 1.1 shows the plan of a bedroom and part of the main room of a house. Other rooms are not shown.

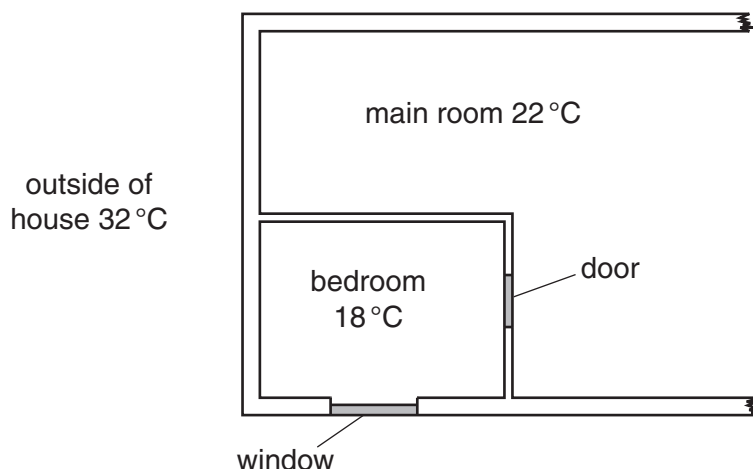


Fig. 1.1

The temperatures of the main room, the bedroom and the outside of the house are shown on Fig. 1.1.

Fig. 1.2 shows all the thermal energy (heat) inputs to the bedroom in one hour.

thermal energy input to bedroom	
through door and walls from main room	50 000 J
through walls from outside of house	2 000 000 J
through window	1 000 000 J
from person sleeping in bedroom	250 000 J

Fig. 1.2

- (a) Suggest why more thermal energy enters the bedroom from the outside of the house than from the main room.

Temperature difference b/w bedroom & outside is greater so heat transfer is greater.[1]

- (b) An air conditioner keeps the temperature constant in the bedroom by removing thermal energy.

- (i) Calculate the total thermal energy that the air conditioner removes from the bedroom in one hour.

$$50,000 + 20,00000 + 10,00000 + 250,000$$

thermal energy = $33 \times 10^5 \text{ J}$ [1]

- (ii) The electrical power input to the air conditioner is 300 W.
Calculate the electrical energy input into the air conditioner in 1 hour.

$$E = Pt = \frac{300}{1000} \times 1$$

$$E = Pt \\ = 300 \times 3600$$

energy = 0.3 kWh or 1.08×10^6 J [2]

- (c) The air conditioner cools the air at the top of the room. This causes a convection current in the room.

Explain how the cold air gives rise to the convection current.

Cold air becomes denser sinks to the bottom while hot air rises to air conditioner. This process continues until thermal equilibrium is established. [3]

[MJ2012/P22/Q3]

- 2 (a) No thermal energy (heat) is transferred from the surface of the Sun to the Earth by either conduction or convection.

Explain why this is so.

Both conduction & convection requires medium & there is no medium b/w sun & earth. [2]

- (b) In a certain country, the climate is very sunny and hot during the day and extremely cold during the night.

Explain how painting the houses white helps to maintain a comfortable temperature both during the day and during the night.

during the day: During the day, heat energy from outer walls reflected back to surrounding.

during the night: During the night, heat energy from inner walls reflected back to house.

[3]

[ON2012/P22/Q5]

- 3 Fig. 3.1 shows food being cooked in an electric grill.

red-hot heating elements

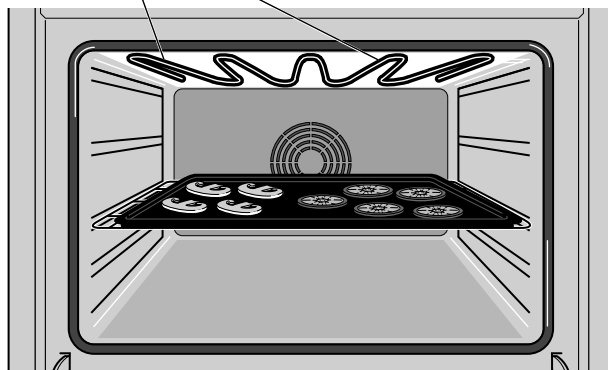


Fig. 3.1

There are red-hot heating elements above the food and thermal energy (heat) is transmitted to the food by radiation.

- (a) Explain what is meant, in this case, by *radiation*.

Mode of transfer of energy without any aid of the medium.
(Infra red).

[2]

- (b) Explain why very little thermal energy is transmitted to the food by

- (i) conduction,

Conduction requires physical contact with the grills while food is not making contact with grills.

[1]

- (ii) convection.

Convection doesn't happen from top to bottom.

[1]

[ON2013/P21/Q4]

- 4 Fig. 4.1 shows a black car going up a hill on a bright, sunny day.

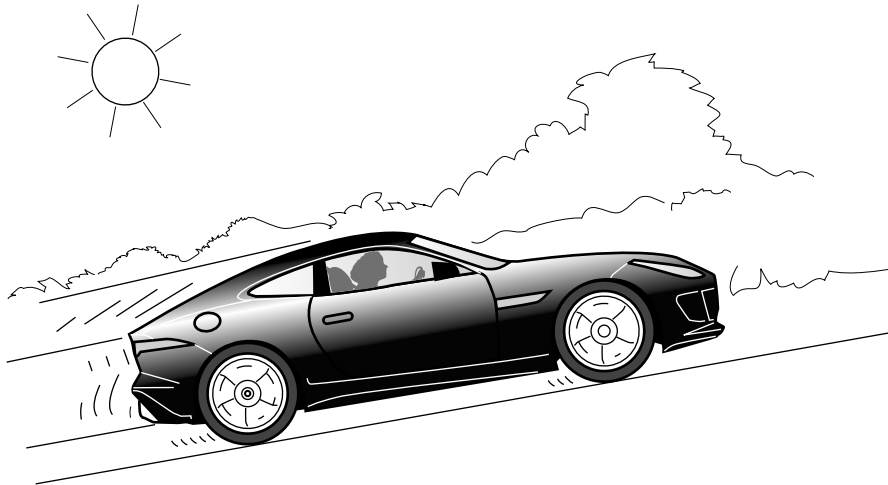


Fig. 4.1

(a) State

- (i) one way in which the car is gaining thermal energy,

By radiation as black is the good absorber of heat.

[1]

- (ii) one way in which the car is losing thermal energy.

By radiation as black is the good emitter of heat.

[1]

- (b) The car accelerates up the hill. In addition to the changes in the thermal energy of the car, there are other energy changes taking place.

Describe the other energy changes occurring as the car moves.

Fossil Fuel Energy is converting into kinetic energy, gravitational energy & sound energy.

[3]

[ON2014/P22/Q4]

- 5 A copper saucepan with a wooden handle contains cold water. The saucepan is placed on a red-hot heating element that is a part of an electric cooker.

Fig. 5.1 shows the saucepan on the heating element.

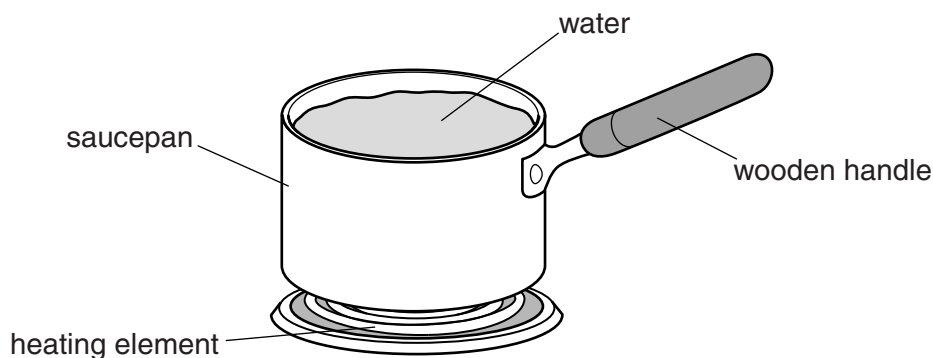


Fig. 5.1

- (a) Explain why wood is a suitable material for the handle of the saucepan.

..... [1]

- (b) (i) Describe and explain, in terms of free electrons, how thermal energy is transferred through the copper base of the saucepan.

.....
.....
.....
.....
.....
..... [3]

- (ii) Describe and explain how thermal energy is transferred upwards through the water.

.....
.....
.....
.....
.....
..... [3]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	22

- 3 (a) large(r) temperature difference (between bedroom and outside)
OR outside is hot(ter than main room) B1
- (b) (i) 3 300 000 J(/hour) B1
- (ii) ($E = P \times t$) in any form; $300 \times 60 \times 60$ C1
 1.08×10^6 J; 1.1×10^6 J
OR 0.3 kWh A1
- (c) cold air sinks B1
(cold air has a) high(er) density or contracts B1
hot air rises
OR hot air has a low(er) density
OR (hot) air comes in to replace cold air B1 [7]

Page 2	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2012	5054	22

- 5 (a) space is a vacuum/empty B1
these methods need matter/medium/molecules
or do not occur in vacuum B1
- (b) any three of:
day: white is a poor absorber/good reflector
day: less heat absorbed/less heating (of house)
night: white is a poor emitter/radiator
night: less heat emitted/heat loss (from house)
anywhere: of IR/radiation/radiant heat B3 [5]

Page 2	Mark Scheme	Syllabus	Paper
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- 4 (a) (energy transmitted) by electromagnetic/infra-red (wave)/can travel
through a vacuum B1
infra-red or visible $< \lambda <$ microwaves or λ just longer than visible
(i.e. infra-red scores 2/2) B1
- (b) (i) air is a poor conductor B1
- (ii) convection occurs (primarily) upwards/hot air rises (**not** heat rises) B1 [4]

Page 2	Mark Scheme	Syllabus	Paper
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- 4 (a) (i) heat gained from burning fuel/combustion or friction between moving
parts/with air/road or from (radiation of) Sun B1
- (ii) heat lost to air/surroundings or by convection (currents) or exhaust/hot
gases/fumes or from exhaust or heat emitted (by hot car) or by radiation B1
- (b) at start chemical energy decreases or at start chemical energy to
gravitational/potential energy (of car) increases or at end of process B1
kinetic energy (of car or air) increases B1 [5]

Page 2	Mark Scheme	Syllabus	Paper
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- 4 (a) wood is a poor/not a conductor or (good) insulator (of heat) B1
- (b) (i) vibrating atoms/ions/particles/molecules or electrons gain energy B1
atoms/ions/particles/molecules hit free electrons or electrons travel (a long
distance through the copper/saucepan) B1
electrons hit/transfer energy to (distant) atoms/ions/molecules/particles B1
- (ii) hot/heated water expands/is less dense B1
hot/heated water/less dense water rises B1
(sets up) circulation/convection (current) or cold water sinks B1 [7]